

would feel each beat of your heart and be constantly aware of every bit of clothing on your body. Furthermore, adaptation is essential for you to see, hear, and smell changes in external stimuli that vary widely in intensity.

Types of Sensory Receptors

Sensory receptors fall into five categories based on the nature of the stimuli they transduce: mechanoreceptors, chemoreceptors, electromagnetic receptors, thermoreceptors, and pain receptors.

Mechanoreceptors

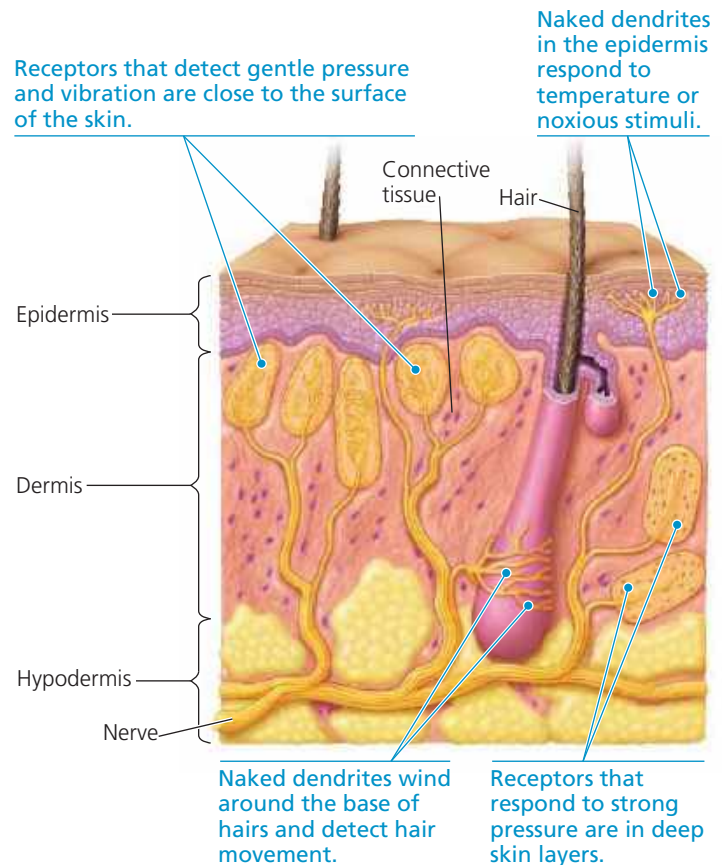
Our senses of hearing and balance, as well as our responses to pressure, touch, stretch, and motion, rely on sensory receptors called **mechanoreceptors**, which sense physical deformation caused by forms of mechanical energy. Mechanoreceptors typically consist of ion channels that are linked to structures that extend outside the cell, such as “hairs” (cilia), and also anchored to internal cell structures, such as the cytoskeleton. Bending or stretching of the external structure generates tension that alters ion channel permeability. This change in turn alters the membrane potential, resulting in a receptor potential—a depolarization or hyperpolarization.

The vertebrate stretch receptor, a mechanoreceptor that detects muscle movement, triggers the familiar knee-jerk reflex (see Figure 49.5). Stretch receptors in vertebrates are dendrites of sensory neurons that spiral around the middle of certain small skeletal muscle fibers. When the muscle fibers are stretched, the sensory neurons depolarize, triggering nerve impulses that reach the spinal cord, activate motor neurons, and generate a reflex response.

Mechanoreceptors that are the dendrites of sensory neurons are also responsible for the mammalian sense of touch. Touch receptors are often embedded in layers of connective tissue. The structure of the connective tissue and the location of the receptors dramatically affect the type of mechanical energy (light touch, vibration, or strong pressure) that best stimulates them (**Figure 50.5**). Receptors that detect a light touch or vibration are close to the surface of the skin; they transduce very slight inputs of mechanical energy into receptor potentials. Receptors that respond to stronger pressure and vibrations are in deep skin layers.

Some animals use mechanoreceptors to literally get a feel for their environment. For example, cats and many rodents have extremely sensitive mechanoreceptors at the base of their whiskers. Like the appendages on the face of the star-nosed mole, whiskers act as touch organs. Deflection of different whiskers triggers action potentials that reach different cells in the brain. As a result, an animal’s whiskers enable the brain to assemble a “touch map” detailing the location of nearby objects such as food or obstacles.

▼ **Figure 50.5 Sensory receptors in human skin.** Most receptors in the dermis are encapsulated by connective tissue. Receptors in the epidermis are naked dendrites, as are hair movement receptors that wind around the base of hairs in the dermis.



Chemoreceptors

Chemoreceptors that monitor the internal environment fall into two broad categories. Some transmit information about overall solute concentration. For example, osmoreceptors in the mammalian brain detect changes in the total solute concentration of the blood and stimulate thirst when osmolarity increases (see Figure 44.19). Other chemoreceptors respond to specific molecules in body fluids, including glucose, oxygen, carbon dioxide, and amino acids.

Animals also use chemoreceptors to detect stimuli in their diet and in the environment they occupy. The antennae of the male silkworm moth contain two of the most sensitive and specific chemoreceptors known (**Figure 50.6**); these receptors can detect components of the sex pheromone released by a female moth several kilometers away. For pheromones and other molecules detected by chemoreceptors, the stimulus molecule binds to the specific receptor on the membrane of the sensory cell and initiates changes in ion permeability.

Electromagnetic Receptors

An **electromagnetic receptor** detects a form of electromagnetic energy, such as light, electricity, and magnetism. For instance, the platypus has electroreceptors on its bill